

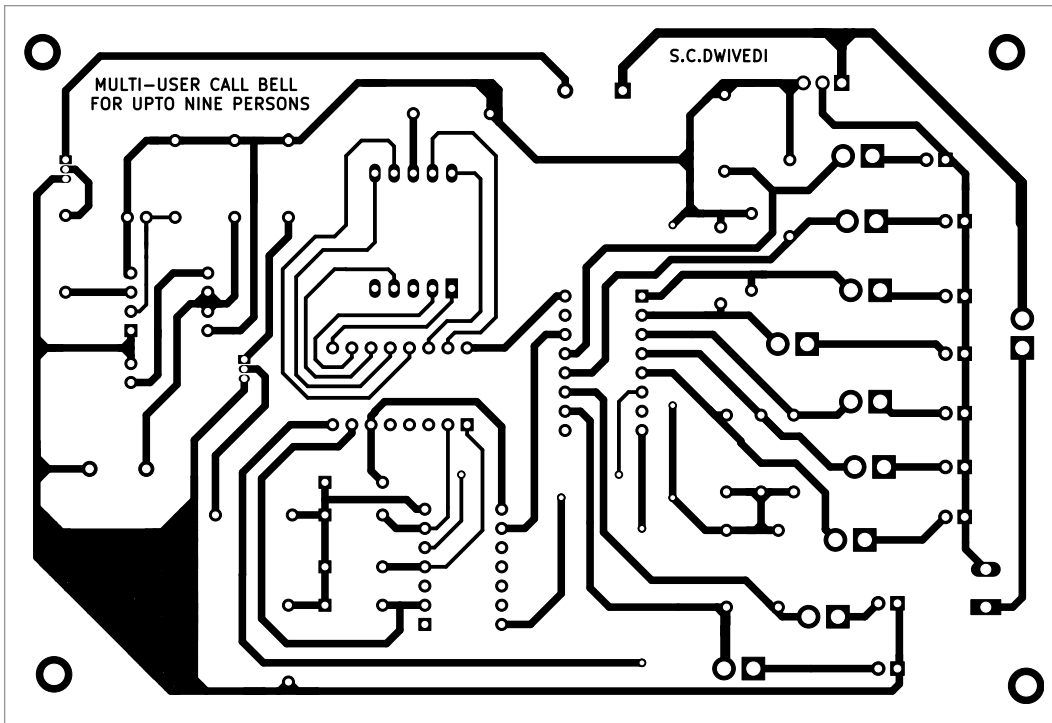


Fig. 4: Actual-size PCB

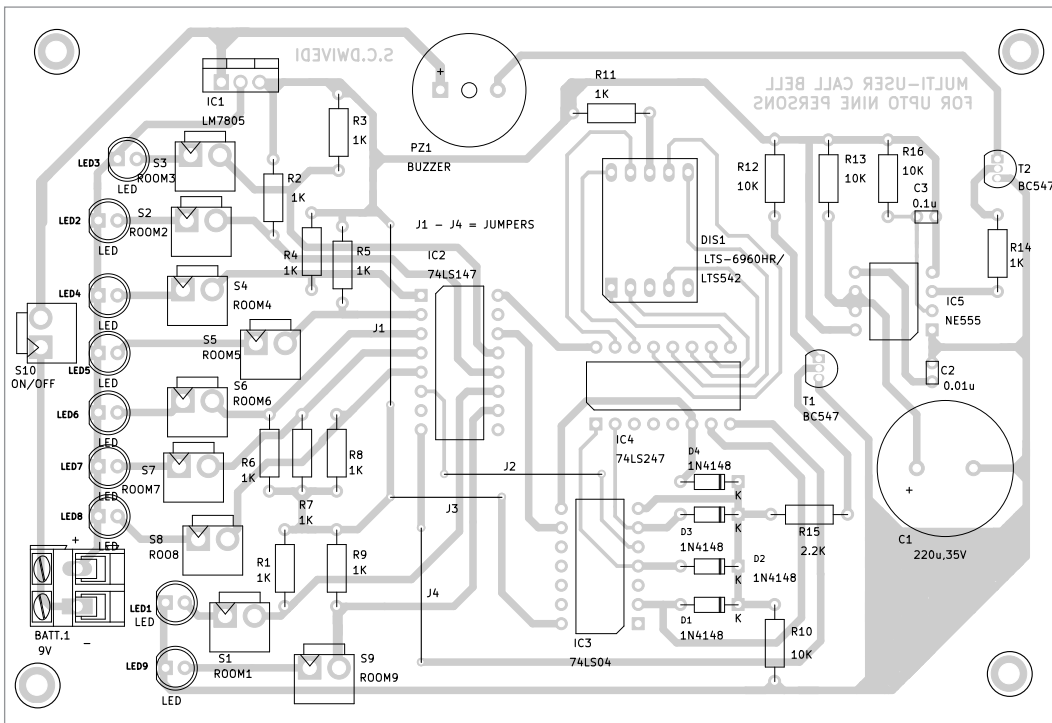


Fig. 5: Component layout of the PCB

S9) are installed at nine different places or rooms along with their corresponding LEDs (LED1 through LED9) for caller indications. IC2

encodes nine data lines into four BCD outputs.

When switches S1 through S9 are open and no LED is glowing,

the seven-segment display DIS1 shows 0. If any of the switches is closed, the BCD output goes low and is inverted by IC3(A) through IC3(D). Then that room number LED glows, and at the same time the seven-segment display DIS1 shows that room number.

Working of the circuit is simple. When, say, room number 1's switch S1 is turned on, its LED1 glows and DIS1 displays 1 besides sounding an alarm through piezo buzzer PZ1. The caller should switch off S1 after the call has been attended so that the system is ready to receive the next call.

An actual-size, single-side PCB layout for the call bell is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Install switches S1 through S9 along with their respective LEDs (LED1 through LED9) in nine different rooms or locations. The assembled PCB, including DIS1 and speaker, should be placed where the

attendant is stationed. **EFY**

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